Whenever you speak to robotics enthusiasts in India, they are sure to mention one or all of the following problems that stand in their way: the dearth of low-cost components, the lack of affordable but high-quality finished products, insufficient knowledge to identify the right components for an application to put them together efficiently, scarce support for complex products, and so on. We set out to explore whether open source hardware could solve some or all of these problems, and were in for a pleasant surprise. Several members of the community felt it could certainly help, not only to sort out the existing problems, but also to improve the robotics scene in India in many other ways.

Problems aplenty

The robotics community faces different problems at different levels. Students, for example, seem to be affected by the unavailability of components. Praveen Pitchai is a member of RoboMSR, a group of robotics enthusiasts who are being mentored by Dr K G Srinivasa at the M S Ramaiah Institute of Technology, Bengaluru. He says, "One of the major hurdles in the field of robotics in India is the lack of components. It is difficult to compete with the frontrunners in the field due to this very reason. Once a new component like a sensor or a micro-controller is found and acquired from vendors outside India, it becomes difficult to configure it to the required specifications within a project, as there are already set standards to be followed by the company that manufactures them. This not only increases the costs involved, but also the complexity."

Rohit de Sa, nucleus member, Centre for Robotics and Intelligent Systems, BITS Pilani, adds, "Components in raw form (resistors, capacitors, chips) are available. However, good-quality finished products (for example, a microcontroller development board, or a long-range radio frequency transceiver) are difficult to find."

On the other hand, Fahad Azad, managing partner of Robosoft Systems, feels that the lack of components by itself is not the main problem. "It has never been only the lack of components in most cases, but also the inability to identify the components required in an application. Even when the components are identified, they are out of reach due to economic factors; thus, the *jugad* mode kicks in," he says.

Open source could be the panacea

Know more about open source

Introduction to open source, open licences and open standards
http://www.opensource.org
Comparison of open and closed source
http://en.wikipedia.org/wiki/Comparison_of_open_source_and_closed_source

All of those contacted agree that open source hardware (OSH) or open hardware can, in some way or the other, sort out these problems. Before looking at how this could be so, we need to understand what OSH really is; it is basically the concept of open source (as popularised by open source software) applied to the design and development of hardware components. Open hardware is not free-of-cost hardware! It merely refers to hardware wherein the underlying technology or building blocks—the design and software—are also available to the user. That implies that design elements such as mechanical drawings, schematics, bill of materials, printed circuit board (PCB) layout data, hardware design language (HDL) source code and integrated circuit (IC) layout data, along with the software drivers and libraries, are all openly published and shared with others under open source licences. This enables engineers to use the hardware better, to embed it in other products, to customise it to their specific needs, manufacture modified or enhanced components, improve the design and share it back with the community of users and developers, and so on, as permitted by the licences used. This model of hardware design and development appears to offer several benefits for the robotics community.

"Researchers have long been used to cranking out code in the morning and having a working prototype by the afternoon, but have been frustrated that they can't do the same with hardware. That's starting to change, and fast, driven in part by robotics enthusiasts and do-it-yourself types who are utilising a new generation of OSH platforms and rapid fabrication tools. We are starting to realise that we can control a laser cutter like a printer, and also a precision range-finder. That means the promise of creating new physical devices as quickly as Web apps are written, is just around the corner," says Shekar N H, another RoboMSR member.

Shekar is inspired by Steve Cousins, a robotics researcher at Stanford, who is using his start-up, Willow Garage, to show the world how open source development can help make low-cost personal robots. Shekar enthusiastically describes more of Cousins' dreams: "Cousins' company plans to make 10 robots running open source code and make them available to researchers around the country in an attempt to finally bring the world into the 'Jetsons' age, where a robot can mop the floor, empty the dishwasher, and even fetch and open a bottle of beer. OSH, complete with diagrams and self-assembly kits, is growing even faster than expected, because of these reasons and the opportunities created."

Let us look at just some of the ways in which open source could help advance developments in the robotics domain.

Reduced cost and time-to-market: "While open source software has helped in developing numerous innovative products and gadgets, in most cases, it has now reached a bottleneck due to the lack of similar platforms for